LAB 5
MEASURING A REFRACTIVE INDEX

BY DAKSH KHANDELWAL

OBJECTIVE AND EXPERIMENT

DETERMINE THE REFRACTIVE INDEX OF THE SOLID POLYETHYLENE DIELECTRIC INSIDE RG-58 COAXIAL CABLE AND CHARACTERIZE HOW IT VARIES WITH SIGNAL FREQUENCY BY MEASURING THE PROPAGATION DELAY OF PULSES THROUGH CABLES OF KNOWN LENGTHS.

- **PRINCIPLE:** $n = \frac{c}{v}$; MEASURE v FROM PULSE TRAVEL TIME THROUGH A KNOWN CABLE LENGTH
- **SETUP:** WAVEFORM GENERATOR SENDS A PULSE INTO RG-58; OSCILLOSCOPE CAPTURES SENT AND REFLECTED/RECEIVED SIGNALS
- **EQUIPMENT:** TEKTRONIX TBS1000B SCOPE, B&K 4010A/4011A GENERATOR, FOUR COAX CABLES OF VARYING LENGTH, 3 BNC TEES, 2 50-Ω TERMINATORS
- **METHOD:** COMPARE DELAYS ACROSS DIFFERENT CABLE LENGTHS TO EXTRACT v, THEN SWEEP FREQUENCY TO TEST FOR DISPERSION
- **EXPECTED:** $n \approx 1.5$ FOR POLYETHYLENE (VELOCITY FACTOR ≈ 0.66)

PHYSICS AND METHOD

Physics

- Phase velocity in a dielectric: $v = 1/\sqrt{(\varepsilon\mu)}$; in vacuum $c = 1/\sqrt{(\varepsilon_0\,\mu_0)}$
- Refractive index: $n = \frac{c}{v}$

Two-cable method

- Send a pulse through a short and a long cable; extra time covers the extra length
- $\Delta t = t_{long} - t_{short}, \Delta d = l_{long} - l_{short}$
- Working equation: $n = \frac{c\,\Delta t}{\Delta d}$
- Uncertainty propagation: $\sigma_n = \sqrt{\left(\frac{c}{\Delta d}\sigma_{\Delta t}\right)^2 + \left(\frac{c\Delta t}{\Delta d^2}\sigma_{\Delta d}\right)^2}$

DATA AND MEASUREMENTS

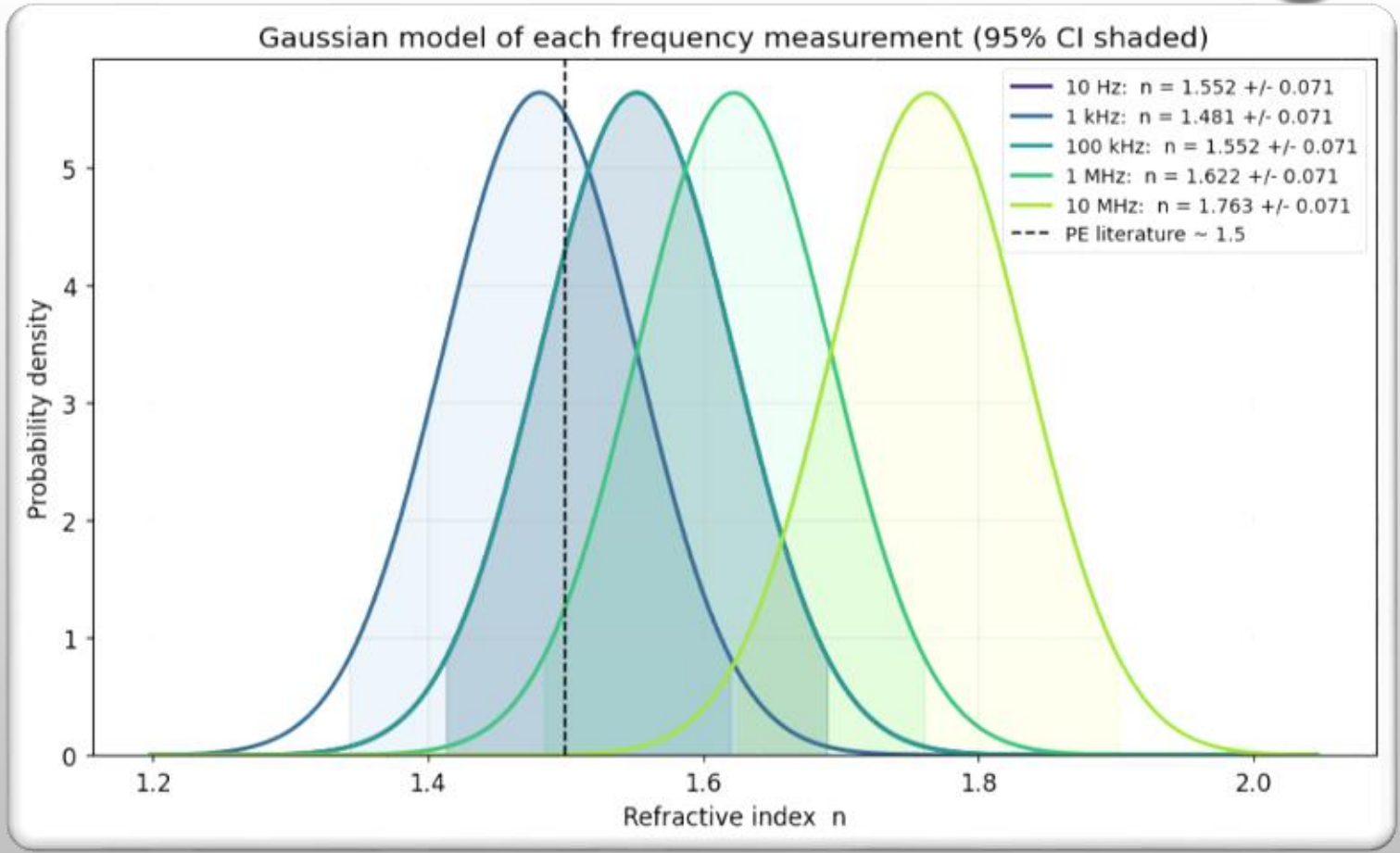
Cable lengths: $L_{short} = 185 \pm 1\ \text{cm}$, $L_{long} = 610 \pm 1\ \text{cm} \rightarrow \Delta d = 425\ \text{cm}\ (\sigma_{\Delta d} = \sqrt{2}\ \text{cm})$
Oscilloscope: Cursor uncertainty $\sigma_{\Delta t} = 1\ \text{ns}$

Frequency	$\Delta t\ (ns)$	n	95% CI
10 Hz	22	1.552	± 0.139
1 kHz	21	1.481	± 0.139
100 kHz	22	1.552	± 0.139
1 MHz	23	1.622	± 0.139
10 MHz	25	1.763	± 0.139

Each σ_n *is dominated by the* ± 1 *ns* Δt *read uncertainty (>99% of variance). The* $\sigma_{\Delta d}$ *contribution is negligible at this* Δd.

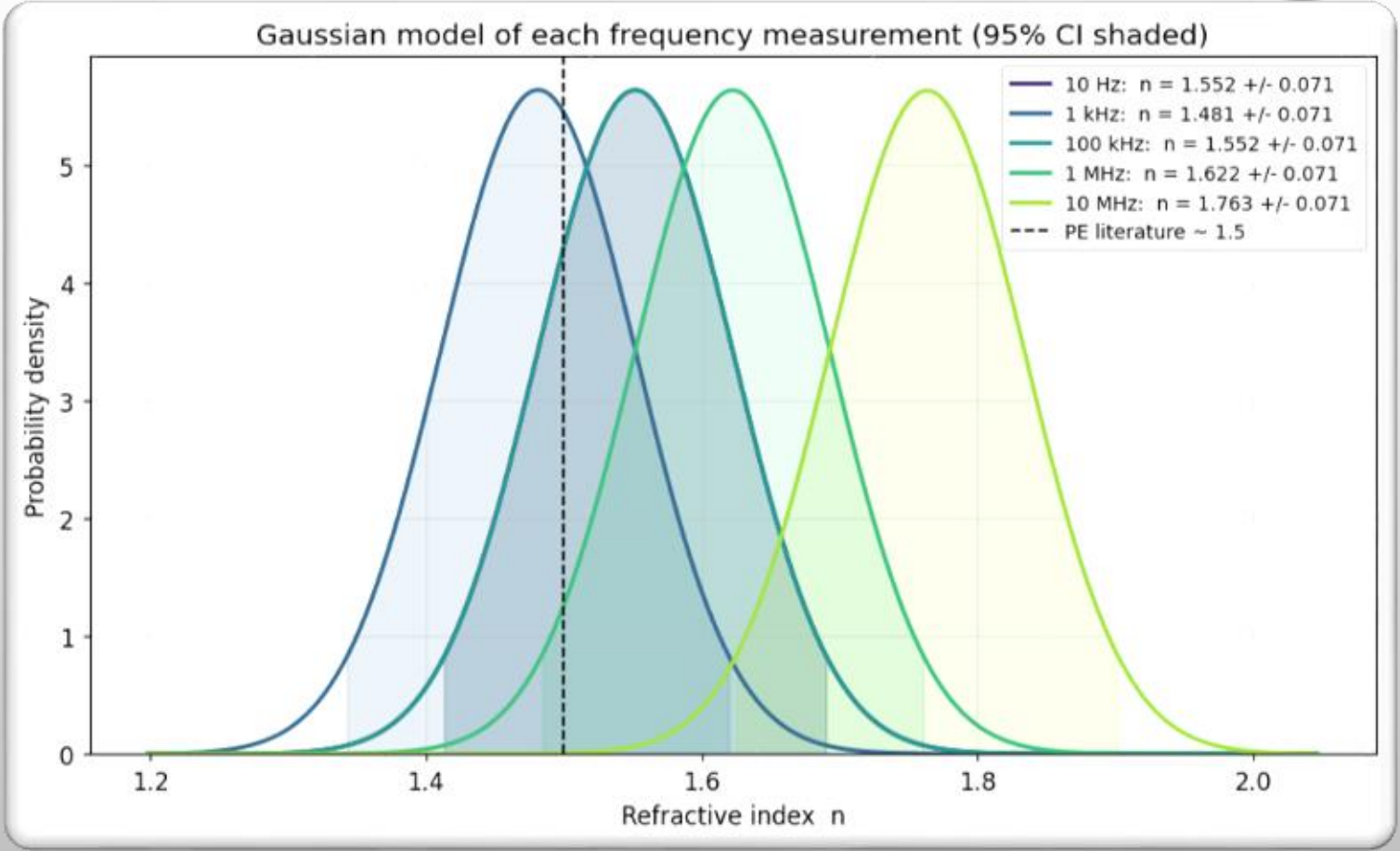
RESULTS: PER-FREQUENCY MEASUREMENTS

- Each measurement modeled as $n(n_i, \sigma_i^2)$: a gaussian centered on n_i with width σ_n
- Shaded band is the $\pm 1.96\ \sigma$ interval — 95% probability the true n lies inside it
- All five σ are nearly identical, so the curves have the same width and only shift horizontally
- Only 4 curves visible: 10 hz and 100 khz both have $\delta t = 22\ ns$, so their pdfs coincide



RESULTS: INTERPRETATION

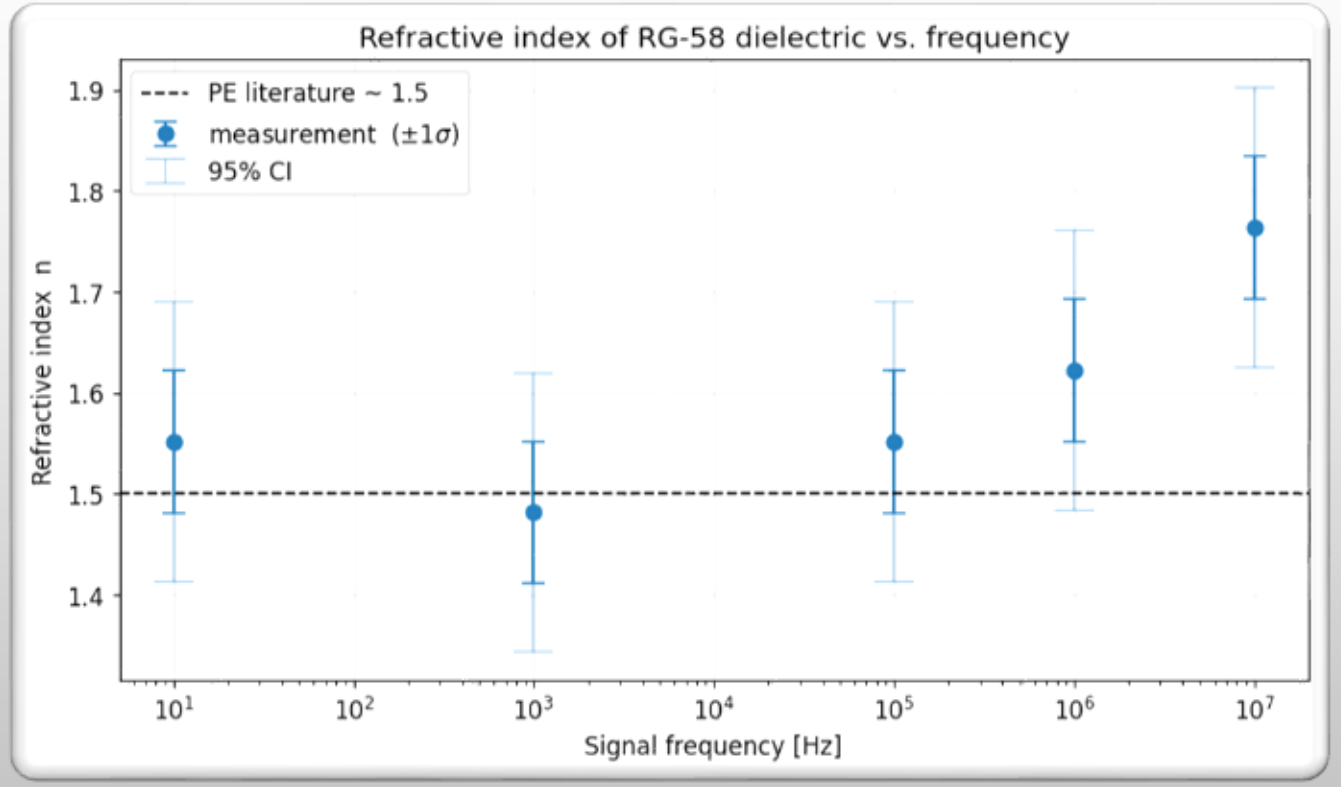
- Heavy overlap shows measurements are statistically consistent with each other
- Dashed line at n = 1.5 (polyethylene) lies inside the 95% band for 10 Hz, 1 kHz, 100 kHz, and 1 MHz
- 10 MHz is the outlier: $n = 1.76 \pm 0.07$. PE line is 3.7σ away, outside the 95% band
- Centers drift right with frequency (1.55 → 1.76); at low frequencies the drift is inside the band, but at 10 MHz it is not
- Curve width is set by the ±1 ns cursor uncertainty, not by the dielectric itself
- Narrower PDFs would require a longer cable or finer cursor placement



Results: n vs frequency

N vs. Frequency

- Flat from 10 hz to 100 khz ($n \approx 1.48 - 1.55$)
- Rises at 1 mhz (1.62) and 10 mhz (1.76)
- Drift is at the edge of the ± 1 ns resolution floor, marginal significance



WEIGHTED MEAN

A weighted mean combines several measurements of the same quantity into one best estimate, giving more influence to points with smaller uncertainty. Each point is weighted by $1/\sigma^2$, so a precise measurement (small σ) counts more than a noisy one. When every point has the same σ, it reduces to the ordinary average.

Formula:

$$\bar{n} = \frac{\sum_i n_i / \sigma_i^2}{\sum_i 1/\sigma_i^2} \qquad \sigma_{\bar{n}} = \frac{1}{\sqrt{\sum_i 1/\sigma_i^2}}$$

Applied to this experiment:

- Five per-frequency measurements: $n_\mathrm{i} = 1.552, 1.481, 1.552, 1.622, 1.763$, each with $\sigma \approx 0.0707$
- Statistical σ (independent δt part) shrinks by $\sqrt{N}$: $0.0707/\sqrt{5} = 0.032$
- Systematic σ (shared δd part) does not shrink: 0.005
- Result: $\bar{n} = 1.594 \pm 0.032\ (1\sigma) = 1.594 \pm 0.063\ (95\%\ \mathrm{ci})$

Conclusions

Key findings

- Measured refractive index of RG-58 polyethylene dielectric: $n = 1.594 \pm 0.032$
- Majority of per-frequency 95% CI brackets the literature value $n \approx 1.5$; the weighted mean sits 2.9σ high, consistent with a small systematic bias (e.G. Cursor placement, connector delays not subtracted)
- Implied velocity factor: $v/c = 1/n \approx 0.63$ (RG-58 spec ≈ 0.66)
- Evidence for frequency dispersion: the 10 MHz point (n = 1.76 ± 0.07) sits 3.7σ above the polyethylene literature value, outside its own 95% CI — it is not a noise fluctuation
- Physically plausible: at 10 MHz the skin depth in copper is tens of $\mu\mathrm{m}$, comparable to the inner conductor radius, and dielectric loss in PE begins to rise. Both cause real pulse-velocity shifts in RG-58 at high f
- Caveat: a single high-f point cannot prove dispersion; more measurements between 1 and 50 MHz are needed to confirm the trend

Redoing the Experiment

- Shrink $\sigma_{\delta t}$ — finer cursor placement, averaging many acquisitions, or a longer cable so $\delta t \gg 1\ \mathrm{ns}$
- Δt uncertainty drives >99% of σ_n — reducing it is the single biggest lever